

An Intelligent Lower Limb Exoskeleton for SCI and Stroke Patients

ABSTRACT

Individuals suffering from Spinal Cord Injury (SCI) and stroke often experience significant mobility limitations and muscle weakness, which severely impact their ability to perform essential activities of daily living. Conventional rehabilitation methods can be time-consuming and labour-intensive for both patients and healthcare professionals. Our project seeks to address these challenges by offering an efficient and effective platform for lower limb therapy and exercise. This topic introduces an innovative solution designed to enhance the rehabilitation and quality of life for individuals affected by SCI and stroke. The "An Intelligent Lower Limb Exoskeleton for SCI and Stroke Patients" is a groundbreaking medical device that combines advanced robotics and intelligent control systems to provide a holistic approach to lower limb rehabilitation. The system is equipped with specially designed mechanisms and attachments to target the ankle and leg muscles effectively. We have planned to provide feedback mechanism to reset the target angle while doing exercise. If patient feel pain at certain angle during exercise they can reset the angle to that position and slowly increment the angle. Patients can perform a wide range of exercises to improve strength, flexibility and range of motion in these areas.

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1. INTRODUCTION

Muscle weakness and spasticity have significantly impacted activities of daily living (ADL). Several conditions have lead to such effects, including stroke and spinal cord injury (SCI). Lower limb disorders limited the motions of patients being conditioned with a specific defect, leading to an increase in their dependency and restrict their motions to a limited degree. Traditional rehabilitation methods are time-consuming and labor-intensive for both patients and healthcare professionals. Due to these limitations, rehabilitation robots have been presented and developed over the years. Rehabilitation robotics is considered as a specific branch of biomedical engineering that roles in decreasing and solving the problems related to traditional rehabilitation.

Our device offers a comprehensive solution for lower limb rehabilitation by integrating robotics and intelligent control systems. It consists of specially designed mechanisms and attachments which focuses on the ankle and leg muscles, providing targeted exercises to enhance strength, flexibility, and range of motion. It helps to perform various physiotherapy exercises and helps in gait training. We have planned to provide feedback mechanism to reset the target angle while doing exercise. If patient feel pain at certain angle during exercise they can reset the angle to that position and slowly increment the angle. The device offers a more efficient alternative to traditional rehabilitation methods, potentially reducing the burden on patients and healthcare professionals. The ultimate goal is to enhance the daily living abilities of individuals with SCI and stroke, potentially leading to increased independence and improved quality of life.

2. PROBLEM DEFINITION

2.1 Motive of the Project

It helps in the recovery of joints, typically after surgery. Its main motive is to facilitate and maintain movement in the joint without the patient actively engaging in the motion. This continuous movement helps prevent stiffness, reduces swelling, and promotes healing by gently stretching and flexing the joint.

By keeping the joint in motion within a controlled range, the CPM machine aims to prevent scar tissue formation, improve circulation, and enhance the overall healing process. It's especially beneficial for joints like the knee, shoulder, elbow, and hip, where maintaining movement can be challenging due to pain or immobility after surgery or injury.

2.2 Objective of the Project

The primary objective is to facilitate and enhance the rehabilitation process following surgery or injury to a joint, particularly in the knees, hips. Its specific objectives include:

- **Reducing Stiffness:** One of the main goals is to prevent or minimize stiffness in the joint by continuously moving it through a controlled range of motion.
- **Promoting Healing:** By gently mobilizing the joint, it helps improve circulation, which can aid in the delivery of nutrients and removal of waste products, thereby promoting faster healing.
- **Preventing Adhesions and Scar Tissue:** It can help prevent the formation of adhesions and scar tissue within the joint, which can restrict movement and impair function if not addressed.
- **Improving Joint Mobility:** The repetitive motion provided by the machine helps to maintain and improve joint mobility, even when active movement may be limited due to pain or swelling.

- Managing Pain: Although not its primary function, the gentle movement provided by the CPM machine can help alleviate pain by promoting the release of endorphins and reducing muscle tension.
- Enhancing Patient Compliance: It offers a passive form of rehabilitation, which can be particularly beneficial in the early stages of recovery when active movement may be challenging or contraindicated. This can improve patient compliance with the prescribed rehabilitation protocol.

3. REQUIREMENTS CAPTURE AND ANALYSIS

3.1 LCD Display

This is I2C interface 16x2 LCD display module, a high-quality 2line 16 character LCD module with on-board contrast control adjustment, backlight and I2C communication interface. For Arduino beginners, no more cumbersome and complex LCD driver circuit connection. The real significance advantages of this I2C Serial LCD module will simplify the circuit connection, save some I/O pins on Arduino board, simplified firmware development with widely available Arduino library.

Specifications:

- Compatible with Arduino Board or other controller board with I2Cbus.
- Display Type: Negative white on Blue backlight.
- I2C Address:0x38-0x3F (0x3F default)
- Supply voltage: 5V
- Interface: I2C to 4bits LCD data and control lines.
- Contrast Adjustment: built-in Potentiometer.
- Backlight Control: Firmware or jumper wire.
- Board Size: 80x36 mm



Fig.3.1. *LCD Display*

3.2 Stepper Motor – Nema17

Features:

- Tapped Holes on 4 corners.
- 12V 0.3Amp in bipolar mode.
- Weight : 220 gm
- Dimensions :42 x 42 x 34 mm
- 5 mm Standard Shaft 24 mm length
- Torque 3.1kgcm
- Ideal for micromouse due to its low weight to torque ratio.
- Can be directly driven using Dual Motor Driver (Electronic 5 Amp) and AVR 40 Pin Rapid Robot controller board V2

Usage:

- Connect +ve to 2 Common wire and drive 4 coils with drivers such as ULN2003/ULN2803 for driving in Unipolar mode. This ic's will not work directly with this motor since current capacity is less. However 2 in parallel can drive this motors.
- Connect H-bridge / Chopper Drive to 4 wires to use in Bipolar mode.



Fig.3.2. *Stepper motor*

3.3 Stepper Motor Driver

Features:

- Automatic current decay mode detection/choice.
- Mixed with slow current decay mode.
- The low power dissipation of synchronous rectifier.
- Internal UVLO(ultra voltage lockout).
- Crossover current protection.
- Thermal shutdown circuit.
- Ground fault protection.
- Loading and short circuit protection.
- The optional five-step mode: full, 1/2, 1/4, 1/8 and 1/16.



Fig.3.3. *Stepper Motor Driver*

3.4 Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.



Fig.3.4. *Arduino uno*

3.5 Coupler

Features:

- The flexibility of this coupler removes stress between motor and driving component
- Measures 1" x 0.7" diameter / 25mm x 19mm diameter
- 5mm inner diameter / 8mm inner diameter
- 15 grams Material : Aluminium
- 1Size : 8mm x 8mm (Ød1 x Ød2)



Fig.3.5. *Coupler*

3.6 Variable Resistor

Features:

- 3 Terminals, Single Linear, Suits Most Standard Knobs.
- Top adjustment type and knurled shaft for easy operate in daily use.
- Main used to control electrical devices such as volume controls on audio equipment.
- Knurled shaft with split design for push-on knob.



Fig.3.6. *Variable resistor*

3.7 Keypad

Features of 4x4 Flat Matrix Keypad:

- Compact Design: The 4x4 matrix layout provides a compact and efficient design for inputting data.
- Versatile Use: Suitable for a variety of applications, from security systems to interactive electronics projects.
- Easy Integration: Interfaces easily with microcontrollers and development boards for seamless integration into projects.
- Durable Construction: Built with durable materials to withstand everyday use.
- Multiple Keys: Offers 16 programmable keys, expanding input options for users.



Fig.3.7. *Keypad*

3.8 Software Explanation

The mainly used 2 software are Arduino IDE for microcontroller coding and Fusion 360 for product designing.

- **Arduino IDE:** The Arduino Integrated Development Environment (IDE) is a software platform used to program Arduino microcontrollers. It provides a user-friendly interface for writing, compiling, and uploading code to Arduino boards. With a simple and intuitive layout, the IDE allows users to write code in the Arduino programming language, which is a simplified version of C/C++. Additionally, it offers a range of built-in functions and libraries that make it easy to interface with hardware components and sensors. Overall, the Arduino IDE serves as a powerful tool for both beginners and experienced developers to create interactive projects and prototypes efficiently.
- **Fusion 360:** Fusion 360 is a comprehensive computer-aided design (CAD), computer-aided engineering (CAE), and computer-aided manufacturing (CAM) software developed by Autodesk. It is a cloud-based platform that offers powerful tools for designing 3D models, simulating their functionality, and generating toolpaths for manufacturing processes. With Fusion 360, users can create parametric designs, allowing for easy modification and iteration of designs. Its intuitive interface and extensive feature set make it suitable for a wide range of applications, from product design to mechanical engineering and beyond.

4. DESIGN

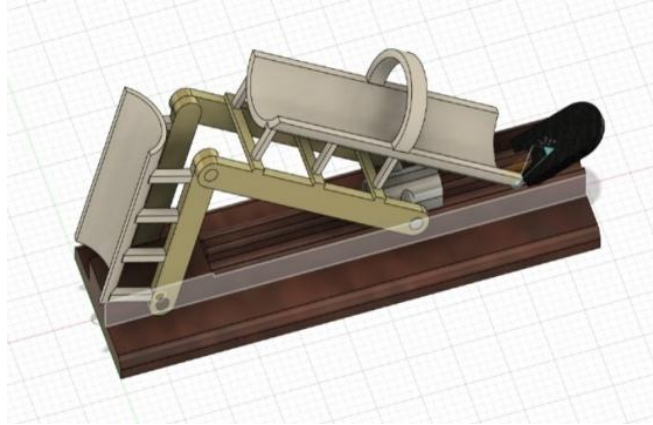


Fig.4.1. *First 3D model of exoskeleton*

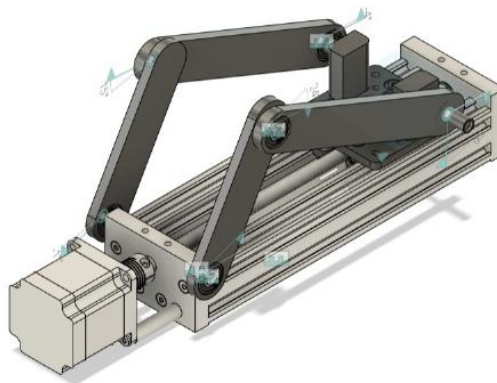


Fig.4.2. *Modified 3D design with accurate measurements*

5. IMPLEMENTATION

5.1 System Architecture of Exoskeleton

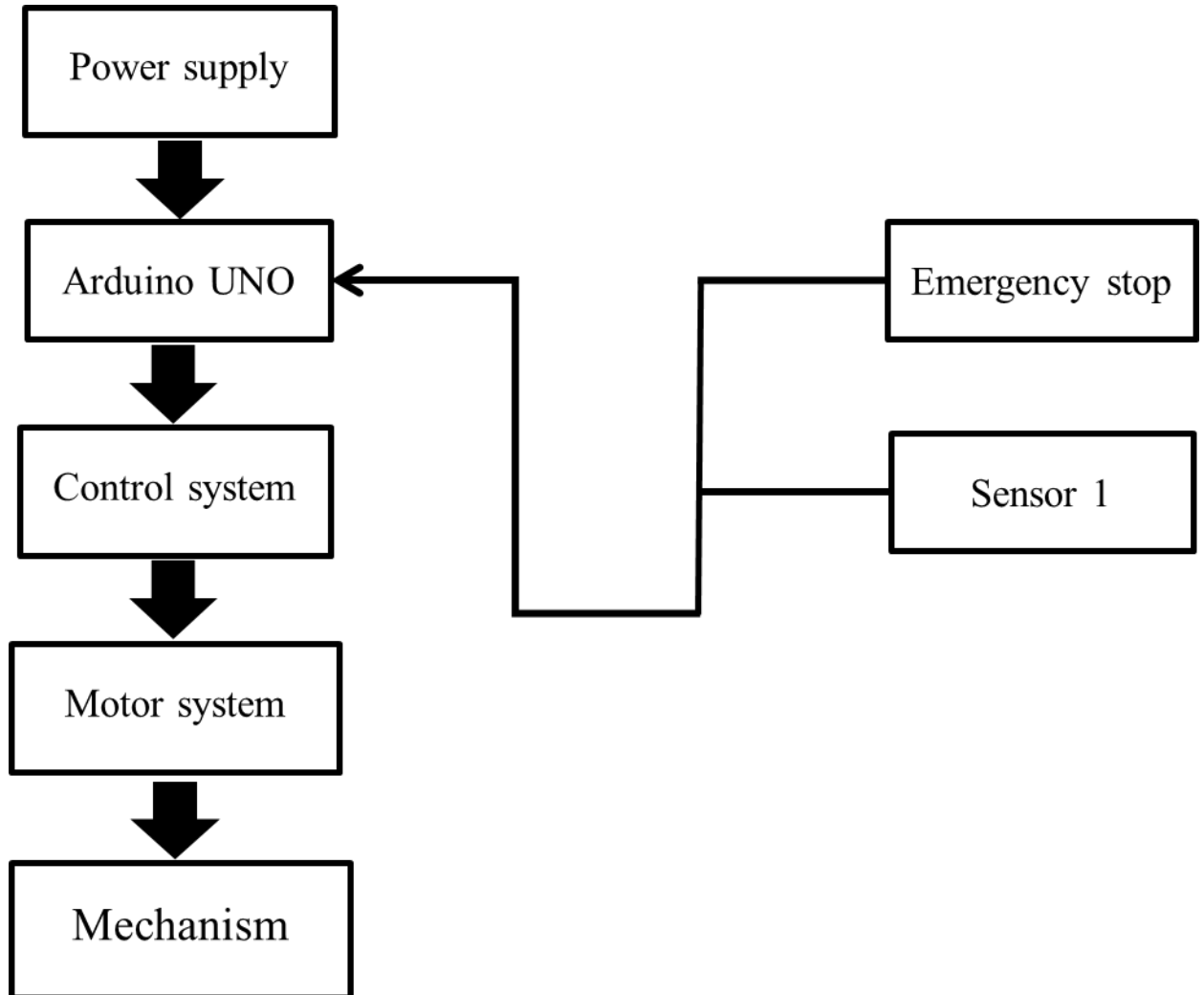


Fig.5.1. *Block Diagram of Exoskeleton*

Here we have a leg holder where the patient can place their leg. Patient can select the target angle along with type of therapy with help of LCD screen and push buttons. If any discomfort is felt by the patient, buttons are provided for stopping the process. The target angle then resets to that previously stopped angle. Patient could increment the target angle gradually.

6. RESULTS AND DISCUSSION

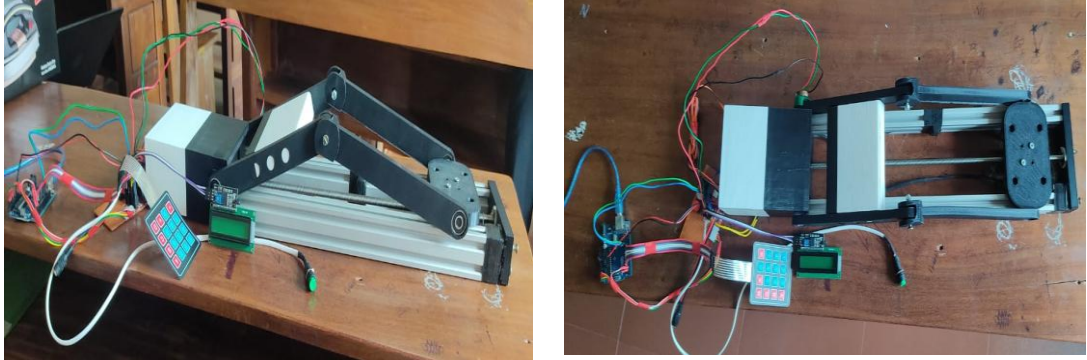


Fig.6.1. *Final product of exoskeleton*

Fig 6.1 depicts the final product. Here we have a leg holder where the patient can place their leg. Patient can select the target angle along with type of therapy with help of LCD screen and push buttons. If any discomfort is felt by the patient, buttons are provided for stopping the process. The target angle then resets to that previously stopped angle. Patient could increment the target angle gradually.

7. CONCLUSION

Lower limb exoskeletons have shown immense potential for improvement of rehabilitation and mobility of individuals with various lower limb disabilities. They offer a very effective means of both assistance and resistance, which can be very specific to individual needs. The studies we have examined consistently show improvements in gait patterns, muscle strength, and overall functional outcomes among patients using these devices. Here we tries to combine both knee and ankle rehabilitation into a single device. This device achieves a closed loop for the rehabilitation process of the stroke patients. We conducted literature survey for about 30 research papers and studied about the various trends in lower limb physiotherapy exoskeletons. We went through various methodologies and technologies and understood the advantages along with its drawbacks. Here we tries to find a solution for that drawbacks. So our device tries to perform basic physiotherapy exercise and provides gait training.

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